

WorkFlow Global HRMS - Part 5: Attendance Module

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1. Introduction

Part 5 implements the Attendance Module for the WorkFlow Global HRMS. This module enables employees to clock in/out, tracks working hours in real-time, and provides managers with team attendance monitoring capabilities with exception filtering.

2. Architecture Overview

2.1 Backend Architecture

The Attendance Module follows the modular monolith pattern with clear separation of concerns:

- Domain Layer: AttendanceRecord entity with status enums
- Infrastructure Layer: Repository pattern for data access
- Application Layer: Business logic and DTOs
- GraphQL Layer: API endpoints for queries and mutations

2.2 Frontend Architecture

The frontend implements a mobile-first responsive design:

- Attendance Page: Clock in/out functionality with live timer
- Live Timer Component: Real-time working hours calculation
- Team Attendance View: Manager dashboard with filtering
- GraphQL Integration: Apollo Client for data fetching

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Generated: June 23, 2026

3. Database Schema

3.1 AttendanceRecord Table

The AttendanceRecord entity stores daily attendance data:

AttendanceRecord:

- Id: string (Primary Key)
- EmployeeId: string (Foreign Key to Employee)
- Date: DateTime (Date of attendance)
- ClockIn: DateTime? (Clock-in timestamp)
- ClockOut: DateTime? (Clock-out timestamp)
- ProductiveHours: decimal (Working hours minus breaks)
- BreakHours: decimal (Break duration, default 1 hour)
- OvertimeHours: decimal (Hours beyond standard shift)
- Status: AttendanceStatus (Present/Late/HalfDay/Absent)
- CreatedAt: DateTime
- UpdatedAt: DateTime

3.2 AttendanceStatus Enum

AttendanceStatus:

- Present: On time (within 15 minutes of shift start)
- Late: Clocked in after grace period
- HalfDay: Less than 4 productive hours
- Absent: No clock-in record

WorkFlow Global HRMS - Part 5: Attendance Module

Generated: June 23, 2026

4. Time Computation Logic

4.1 Clock-In Logic

When an employee clocks in:

- Create new AttendanceRecord for today
- Set ClockIn to current UTC time
- Determine status: Present if within 15 min grace period, else Late
- Initialize ProductiveHours, BreakHours, OvertimeHours to 0

4.2 Clock-Out Logic

When an employee clocks out:

- Update existing record with ClockOut timestamp
- Calculate total hours: ClockOut - ClockIn
- Calculate ProductiveHours: Total - 1 hour lunch break
- Calculate OvertimeHours: If total > 9 hours, overtime = total - 9
- Update status to HalfDay if ProductiveHours < 4

4.3 Calculation Formulas

Standard Shift: 9:00 AM - 6:00 PM (9 hours including 1-hour lunch)

Grace Period: 15 minutes after shift start

Total Hours = ClockOut - ClockIn

Productive Hours = max(0, Total Hours - 1.0)

Overtime Hours = max(0, Total Hours - 9.0)

Status Logic:

- ClockIn > 9:15 AM => Late
- ProductiveHours < 4.0 => HalfDay
- Otherwise => Present

WorkFlow Global HRMS - Part 5: Attendance Module

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5. Live Timer Implementation

5.1 Client-Side Timer

The live timer runs entirely on the client to avoid server load. It calculates elapsed time by comparing the clock-in timestamp with the current time, updating every second.

```
useEffect(() => {
  const calculateElapsedTime = () => {
    const clockIn = new Date(clockInTime);
    const now = new Date();
    const diff = now.getTime() - clockIn.getTime();

    const hours = Math.floor(diff / (1000 * 60 * 60));
    const minutes = Math.floor((diff % (1000 * 60 * 60)) / (1000 * 60));
    const seconds = Math.floor((diff % (1000 * 60)) / 1000);

    return `${hours}:${minutes}:${seconds}`;
  };

  const interval = setInterval(() => {
    setElapsedTime(calculateElapsedTime());
  }, 1000);

  return () => clearInterval(interval);
}, [clockInTime]);
```

5.2 Benefits of Client-Side Timer

- No server polling required - reduces backend load
- Real-time updates without network latency
- Works offline after initial data fetch
- Accurate to the second without API calls

WorkFlow Global HRMS - Part 5: Attendance Module

Generated: June 23, 2026

6. GraphQL API

6.1 Queries

`getTodayAttendance(employeeId: String!): AttendanceDto`

- Returns today's attendance record for an employee
- Used by the attendance page to show current status

`getMyAttendance(
 employeeId: String!,
 startDate: DateTime!,
 endDate: DateTime!`

`): [AttendanceDto!]!`

- Returns attendance history for date range
- Used for the history tab

`getTeamAttendance(
 managerId: String!,
 date: DateTime!,
 statusFilter: AttendanceStatus`

`): [AttendanceDto!]!`

- Returns team attendance for a specific date
- Optional status filter for exception reporting
- Only accessible by Manager, HR, Admin roles

6.2 Mutations

`clockIn(employeeId: String!): AttendanceDto`

- Creates attendance record with clock-in time
- Determines status based on time
- Returns updated attendance record

`clockOut(employeeId: String!): AttendanceDto`

- Updates record with clock-out time
- Calculates productive, break, and overtime hours
- Returns final attendance record

WorkFlow Global HRMS - Part 5: Attendance Module

Generated: June 23, 2026

7. Frontend Features

7.1 Employee Attendance View

The main attendance page provides:

- Clock In button when not clocked in
- Live timer showing working hours when clocked in
- Clock Out button to end the day
- Summary cards showing productive, break, and overtime hours
- Shift schedule information (9 AM - 6 PM)
- History tab with past 30 days of attendance records

7.2 Manager Team View

Managers, HR, and Admins can access the team attendance view:

- Date selector to view any day
- Status filter buttons (All, Present, Late, Half Day, Absent)
- Team member cards with attendance details
- Visual highlighting for exceptions (red for Absent, yellow for Late)
- Quick overview of productive, break, and overtime hours per employee

7.3 Mobile-First Design

The UI is optimized for mobile devices:

- Touch-friendly buttons and controls
- Responsive grid layouts
- Bottom navigation for easy thumb access
- Swipeable tabs for Today/History views
- Dark mode support

WorkFlow Global HRMS - Part 5: Attendance Module

Generated: June 23, 2026

8. Technical Implementation Details

8.1 Repository Pattern

The AttendanceRepository extends IPostgresRepository and provides:

```
GetTodayAttendanceAsync(employeeId, date)
- Fetches single record for employee on specific date
- Uses date comparison for exact match

GetAttendanceHistoryAsync(employeeId, startDate, endDate)
- Returns ordered list of records in date range
- Sorted by date descending

GetTeamAttendanceAsync(employeeIds, date)
- Fetches multiple employee records for one date
- Used by manager dashboard

GetTeamAttendanceRangeAsync(employeeIds, startDate, endDate)
- Bulk fetch for team over date range
- Supports reporting and analytics
```

8.2 Service Layer

AttendanceService implements business logic:

- Validates clock-in/out operations
- Prevents duplicate clock-ins
- Calculates hours and status
- Maps domain entities to DTOs
- Enforces business rules (grace period, half-day threshold)

WorkFlow Global HRMS - Part 5: Attendance Module

Generated: June 23, 2026

9. Interview Preparation Questions

9.1 System Design Questions

Q: How would you handle timezone differences for a global workforce?

A: Store all timestamps in UTC in the database. Convert to employee's local timezone in the frontend based on their profile. Add a Timezone field to Employee entity. Calculate shift times relative to employee timezone.

Q: How would you prevent time theft or buddy punching?

A: Implement biometric verification (fingerprint/face recognition), GPS location tracking with geofencing, IP address logging, and photo capture during clock-in. Add verification fields to AttendanceRecord (Location, IPAddress, PhotoUrl, VerificationMethod).

Q: How would you scale this system for 100,000+ employees?

A: Implement database sharding by company/department, use Redis caching for today's attendance, implement read replicas for reporting queries, use message queues for async processing, and implement CDN for static assets.

9.2 Technical Questions

Q: Why calculate the timer on the client instead of polling the server?

A: Client-side calculation eliminates server load from constant polling, provides real-time updates without network latency, works offline, and scales better. The clock-in timestamp is the source of truth, so calculations are deterministic.

Q: How do you handle clock-out if the employee forgets?

A: Implement auto-clock-out at shift end time, send reminder notifications, allow managers to manually adjust records, and create a pending approval workflow for late clock-outs. Add a IsAutoClockOut flag to track automatic closures.

Q: How would you implement shift patterns (night shifts, rotating shifts)?

A: Create a ShiftPattern entity with start/end times, add ShiftPatternId to Employee, modify status calculation to use employee's shift pattern instead of fixed 9-6, support multiple shift patterns per employee for rotating schedules.

WorkFlow Global HRMS - Part 5: Attendance Module

Generated: June 23, 2026

10. Best Practices Implemented

10.1 Code Quality

- Repository pattern for data access abstraction
- Service layer for business logic separation
- DTO pattern for API contracts
- GraphQL for flexible querying
- TypeScript for type safety

10.2 Performance

- Client-side timer to reduce server load
- Efficient database queries with proper indexing
- Lazy loading for history data
- Optimistic UI updates

10.3 User Experience

- Mobile-first responsive design
- Real-time feedback with live timer
- Clear visual status indicators
- Exception highlighting for managers
- Dark mode support

WorkFlow Global HRMS - Part 5: Attendance Module

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11. Future Enhancements

11.1 Advanced Features

- Biometric authentication integration
- GPS geofencing for location verification
- Shift pattern management
- Overtime approval workflow
- Attendance analytics and reporting
- Integration with payroll for automatic calculations

11.2 Notifications

- Push notifications for clock-in reminders
- Email alerts for missed clock-outs
- Manager notifications for team exceptions
- Weekly attendance summary reports

11.3 Reporting

- Monthly attendance reports
- Overtime trend analysis
- Department-wise attendance statistics
- Export to Excel/PDF
- Custom report builder

WorkFlow Global HRMS - Part 5: Attendance Module

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12. Conclusion

Part 5 successfully implements a comprehensive Attendance Module with clock in/out functionality, real-time working hours tracking, and manager oversight capabilities. The implementation follows best practices with clean architecture, efficient data access, and a mobile-first user interface.

Key achievements include:

- Complete backend implementation with repository and service layers
- GraphQL API with queries and mutations
- Live timer component for real-time hour tracking
- Employee attendance page with clock in/out
- Manager team attendance view with filtering
- Mobile-responsive design with dark mode

The module is production-ready and can be extended with additional features like biometric verification, shift patterns, and advanced reporting as business needs evolve.